

Heuristics and Biases

ED 323: Behavioural Economics

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Heuristic - a mental shortcut

- Heuristics are **mental shortcuts** or **rules of thumb** that people use to make decisions. They **differ from optimization** in that they typically involve a **limited information set** and a **more computationally tractable** decision method.
- The **recognition heuristic** - if one of two objects is recognized, infer that the recognized object is more likely to be the target object.
 - Eg: predict which of two players will win tennis match.

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Heuristic - a mental shortcut

- There is substantial evidence that people use heuristics.
 - People don't normally calculate conditional probabilities using **Bayes' rule**. Instead, the heuristics might approximate Bayes rule under certain conditions.
 - Heuristics are often **accurate and tractable**, but in some environments can **lead to error**.
 - Tversky and Kahneman (1974) defined three classic heuristics:
 - representativeness, availability and anchoring.
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Representativeness heuristic

- Suppose you wish to estimate the probability that an event or person belongs to a certain class.
 - “What is the probability that event A belongs to class B?”
 - “What is the probability that process B will generate event A?”
 - Under the representativeness heuristic, people evaluate probabilities by the degree to which A is representative of (similar to) B.
 - People judge probability by similarity or stereotype matching.
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Representativeness heuristic

- Tversky and Kahneman (1974) provide the following example:
 - An individual who has been described by neighbor as follows:
 - “Steve is very shy and withdrawn, invariably helpful, but with little interest in people, or in the world of reality. A meek and tidy soul, he has a need for order and structure, and a passion for detail.”
 - What is the probability that Steve is a librarian?
 - assessed by the degree to which he is representative of, or similar to, the stereotype of a librarian.
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Representativeness heuristic

- This approach to the judgement of probability leads to **neglect of statistical principles** causing serious errors.
 - Base-rate neglect (insensitivity to prior probabilities of outcomes)
 - Personality description of group of 100 professionals – engineers and lawyers.
 - Group A : 70 engineers and 30 lawyers
 - Group B : 30 engineers and 70 lawyers
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Representativeness heuristic

- This approach to the judgement of probability leads to **neglect of statistical principles** causing serious errors.
 - Sample size neglect (Insensitivity to sample size)
 - A town is served by two hospitals. Large hospital – 45 babies born /day. Small hospital – 15 babies born/day
 - Now, if 50% of all babies are boys. But exact percentage varies from day to day. In one year, each hospital recorded the days on which more than 60% of the babies born were boys. Which hospital you think recorded more such days?
 - Response:
 - The large hospital : 21
 - The small hospital : 21
 - About the same : 53
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Representativeness heuristic

- This approach to the judgement of probability leads to **neglect of statistical principles** causing serious errors.
 - Conjunction fallacy (thinking specific conditions are more probable)
 - In considering tosses of a coin, people regard the sequence
 - H-T-H-T-T-H more likely than the sequence
 - H-H-H-T-T-T which does not appear random, and
 - H-H-H-H-T-H which does not represent the fairness of the coin.
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Availability heuristic

- Under the availability heuristic, people assess the frequency of a class or the probability of an event by the ease with which they can recall instances or occurrences.
 - Assessing risk of heart-attack among middle-aged people by recalling such occurrences among people you know.
 - Probability of success of a business venture by imaging various difficulties it could encounter.
- Availability is a useful clue for assessing frequency or probability because instances of large classes are usually recalled better and faster than instances of less frequent classes.

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Availability heuristic

- Availability is also affected by factors other than frequency and probability. So, complete reliance availability may lead to predictable biases.
 - Biases due to the retrievability of instances.
 - Biases due to the effectiveness of a search set.
 - Biases of imaginability.
 - Illusory correlation.

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Availability heuristic

- In one experiment, Tversky and Kahneman (1974) gave experimental subjects lists of names.
 - In some lists, the men were more famous than the women, and in other lists, vice versa.
 - After viewing the list, they were asked whether the list had more men or women.
- For each list, the subjects judged that the sex with more famous names was more common. Those names were more available in their minds.

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Adjustment and anchoring heuristic

- When using anchoring and adjustment, people estimate by starting from an initial value and adjust from that value to obtain the final estimate.
 - Suppose you know the odds of outcome A, and want to estimate the odds of outcome B. Anchoring and adjustment implies that you will start with the odds of outcome A and adjust to obtain the odds of outcome B.

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Adjustment and anchoring heuristic

- The **accuracy of anchoring and adjustment** depends on the **anchor's quality** and the **size** of the adjustment from the anchor.
 - The **quality of the anchor** relates to the **strength** of the **correlation between the anchor and the quantity** being estimated.
 - Empirically, people tend to use weak or irrelevant anchors.
 - The **size of the adjustment** then needs to **account** for the **relationship between the anchor and the quantity** being estimated.
 - Empirically, observed adjustments from the anchor are too small.
- Example:- Tversky and Kahneman (1974): Percentage of **African countries in the United Nation**.

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The Cab Problem (Tversky & Kahneman, 1982)

- The Scenario:
 - A hit-and-run accident occurred at night involving a **cab**.
 - In the city:
 - 85% of the cabs are **Green**,
 - 15% are **Blue**.
 - A **witness** identifies the **cab as Blue**.
 - The court tests the **witness's accuracy** under similar conditions and finds they are **80%** reliable (i.e., correctly identify color 80% of the time).
- The Question:
 - What is **the probability that the cab involved in the accident was actually Blue**?

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The Cab Problem (Tversky & Kahneman, 1982)

What People Usually Answer

- Most people say “80%”, matching the witness’s reliability.
- They ignore the base-rate information (85% Green, 15% Blue).

Normative (Bayesian) Reasoning

- According to **Bayes’ Theorem**:

$$\begin{aligned}
 P(\text{blue}|\text{claim blue}) &= \frac{P(\text{claim blue}|\text{blue})P(\text{blue})}{P(\text{claim blue})} \\
 &= \frac{P(\text{claim blue}|\text{blue})P(\text{blue})}{\left(P(\text{claim blue}|\text{blue})P(\text{blue}) + P(\text{claim blue}|\neg\text{blue})P(\neg\text{blue}) \right)} \\
 &= \frac{0.8 \times 0.15}{0.8 \times 0.15 + 0.2 \times 0.85} \\
 &= 0.41
 \end{aligned}$$

$$\underbrace{P(\text{claim blue}|\text{blue})}_{80\%} \neq \underbrace{P(\text{blue}|\text{claim blue})}_{\text{Requires Bayes' rule}}$$

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The bias-variance trade-off

- Heuristics, while often associated with errors, can be powerful and effective decision-making tools due to the bias-variance trade-off.
- The bias-variance trade-off suggests that as model complexity increases, bias tends to decrease, but variance (sensitivity to data fluctuations) increases. There’s an optimal level of complexity that balances these two factors.
- Simple heuristics can sometimes outperform more complex decision-making strategies by finding a better bias-variance trade-off, leading to lower overall error despite using less information.
 - The “Take the Best” heuristic
 - The gaze heuristic

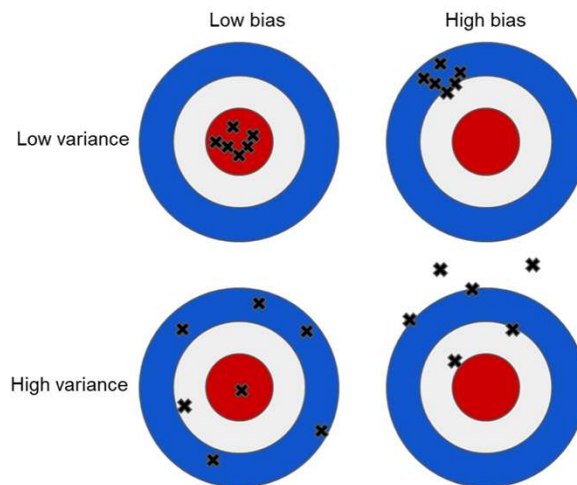
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Understanding bias and variance

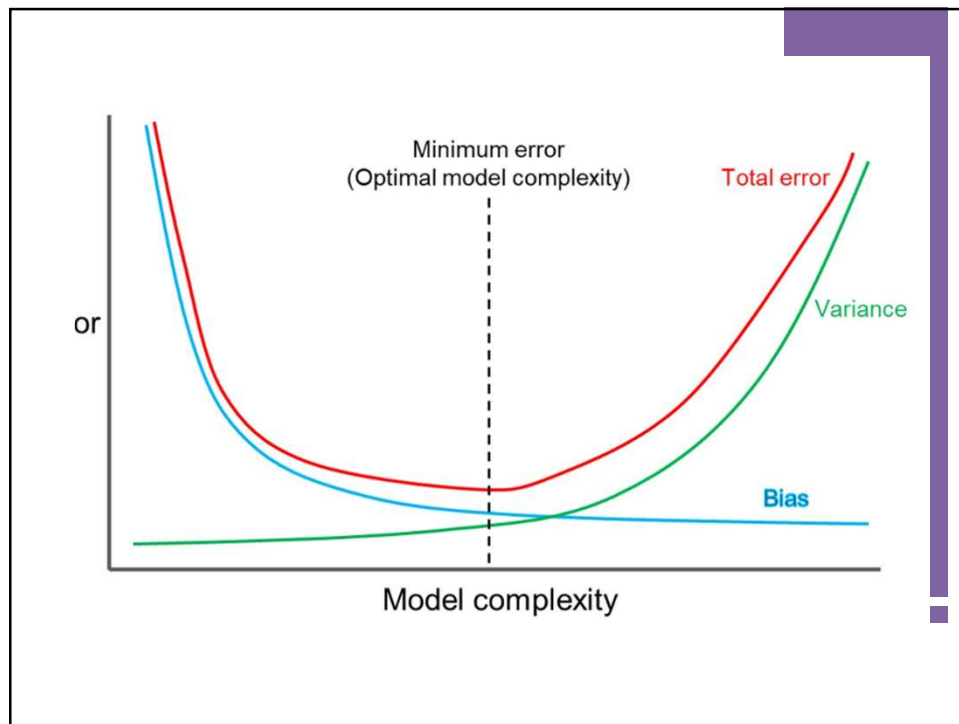
- Bias: -
 - Bias is the degree to which there are erroneous assumptions in your model.
 - The classic case of bias is when you have failed to include a relevant predictor.
- Variance: -
 - The inclusion of too many predictors can lead to what is called variance, which is an error that arises because of the sensitivity of the model to fluctuations in the data you use to develop the model.
 - It ultimately involves giving too much weight to irrelevant or marginally relevant information.
- Predict the level of dropout in a school

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- An unbiased predictor will tend to center on the target. A low variance predictor will tend to cluster. A low variance, low bias estimate is the best outcome.



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Overconfidence

- “Perhaps the most robust finding in the psychology of judgment and choice is that people are overconfident.”

De Bondt and Thaler (1995)

- Overconfidence is a robust finding in judgment and decision-making research, but it actually encompasses three distinct phenomena:
 - Overprecision,
 - Overestimation, and
 - Overplacement.

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Overconfidence

- **Overprecision** is the tendency to believe our predictions or estimates are more accurate than they are, often demonstrated in confidence interval tasks.
- **Overestimation** is the belief that we can perform better than we realistically can, typically occurring for difficult tasks but not for easy ones.
- **Overplacement** is the erroneous judgment that we are better than others, which varies with task difficulty. People tend to over-place on easy tasks and under-place on difficult ones.

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Thank you!

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